Self Introduction

Hi, I'm Srinivasa Rao Talari. I recently completed my B.Tech in Artificial Intelligence and Data Science from Velagapudi Ramakrishna Siddhartha Engineering College. Before that, I completed a diploma in Computer Engineering, which helped me build a strong technical foundation in programming and software development.

I have a strong interest in AI/ML, Python development, and real-world problem solving. I have worked on projects like an XAI-powered insurance claim fraud detection system using XGBoost, SVM, and LLM integration, where I focused on improving model accuracy and reducing false positives.

Apart from academics, I actively spend time learning new technologies, exploring AI tools, and improving my practical skills through online courses and hands-on projects. I enjoy adapting to new technologies and continuously improving myself, which motivates me to grow both technically and professionally.

I'm now looking for an opportunity where I can apply my skills, learn from experienced professionals, and contribute effectively to the organization.

Why should we hire you?

You should hire me because I have strong learning ability, adaptability, and hands-on project experience in AI/ML and software development. During my internship, I initially did not have experience with the Django framework, but I quickly learned it independently and successfully applied it in real-time projects. This shows my ability to adapt to new technologies and contribute effectively to project requirements.

I also worked on projects outside my academic syllabus, such as the Smart Safety Helmet project, where I learned IoT concepts, sensor integration, and real-time communication technologies on my own and implemented them successfully as part of the project.

Along with technical skills, I have good teamwork and time management abilities. I consistently completed internship tasks on time and also handled team coordination responsibilities in projects by conducting discussions and tracking progress regularly.

Since this role requires strong fundamentals in AI/ML, Python, and practical problem-solving, I believe my project experience, quick learning ability, and dedication make me a suitable candidate for this opportunity.

Improved Short Strength Line

- Quick learner with strong adaptability to new technologies and project requirements.
- Strong problem-solving skills with hands-on AI/ML project experience.
- Responsible team player with good communication and time management skills.
- Self-motivated learner who enjoys exploring and implementing new technologies.
- Able to quickly understand concepts and deliver tasks efficiently.

Where do you see yourself in 5 years? (Extended Answer)

In the next five years, I see myself growing into a skilled AI/ML Engineer who can contribute to real-world projects and build impactful solutions using Artificial Intelligence and Machine Learning. I want to continuously improve my technical knowledge in areas like Machine Learning, Deep Learning, Generative AI, and scalable AI systems.

I also want to gain strong industry experience by working on challenging projects, collaborating with experienced professionals, and learning best practices followed in the industry.

Along with technical growth, I would like to take more responsibilities, improve my leadership and communication skills, and eventually contribute as a reliable team member who can guide projects and support junior members as well.

Overall, my goal is to continuously learn, grow professionally, and become a valuable asset to the organization through consistent performance and dedication.

Why did you choose AI and Data Science?

I chose AI and Data Science because I enjoy problem-solving and working with technology. I'm interested in how data and intelligent systems can be used to solve real-world problems and automate tasks efficiently. Through my projects and learning experience, I developed a strong interest in Machine Learning and AI-based applications, which motivated me to build my career in this field.

An Explainable Artificial Intelligence Approach for Health Insurance Claim Fraud Detection

One of my best academic projects is **"An Explainable Artificial Intelligence Approach for Health Insurance Claim Fraud Detection."** I selected this problem because health insurance fraud causes huge financial losses for insurance companies, and traditional manual verification methods are slow, error-prone, and difficult to scale. The main goal of this project was to automate fraud detection, improve claim verification speed, and support transparent decision-making using Machine Learning, NLP, OCR, and Explainable AI techniques.

In this project, I designed and developed a hybrid AI-based fraud detection system using **SVM, XGBoost, OCR, NLP, BioGPT embeddings, SHAP Explainable AI, and GPT-based chatbot integration** to identify whether an insurance claim is fraudulent or legitimate.

Initially, I trained both **SVM and XGBoost** models separately and analyzed their performance. SVM performed well for high-dimensional and text-based medical data, especially for classification tasks. However, it had limitations such as sensitivity to noisy data and slower performance on large datasets. To overcome these limitations, I integrated **XGBoost**, which efficiently handled structured data, feature importance analysis, class imbalance, and scalability. By combining both models, the system achieved approximately **98.48% accuracy** with improved fraud detection performance and balanced prediction reliability.

A major challenge in the project was handling unstructured insurance documents such as scanned medical reports, bills, discharge summaries, and claim forms. To solve this issue, I implemented **OCR (Optical Character Recognition)** to extract machine-readable text from uploaded documents.

After text extraction, I applied **Natural Language Processing (NLP)** techniques such as:

- data cleaning,
- tokenization,
- stop-word removal,
- normalization,
- and feature extraction.

During the preprocessing stage, I used **BioGPT embeddings** to improve contextual understanding of healthcare-related terminology and medical claim descriptions. BioGPT helped convert extracted medical text into meaningful vector representations while preserving semantic relationships between medical terms, treatments, diagnoses, and claim-related information. These contextual embeddings improved

feature quality and enhanced the fraud detection capability of the machine learning models.

Another major challenge was dataset imbalance because fraudulent claims were much fewer than genuine claims. To address this issue, I implemented:

- **SMOTE (Synthetic Minority Oversampling Technique)**
- and random undersampling.

This improved model balance and increased fraud detection capability.

To improve transparency and explainability, I implemented **SHAP (Shapley Additive Explanations)**, which displayed the key parameters responsible for each fraud prediction. This helped admins and auditors understand why a particular claim was flagged as fraudulent and improved trust in the AI system.

Additionally, I integrated a **GPT-powered chatbot** that could answer user queries related to insurance claims in real time. The chatbot explained claim status, fraud analysis results, and policy-related information, improving transparency and user interaction. I also used web scraping techniques to retrieve updated policy-related information dynamically.

I built a **Flask-based full-stack web application** around the AI pipeline. The system included:

- secure email-based login and signup,
- role-based dashboards,
- claim document upload functionality,
- fraud analysis reports,
- chatbot integration,
- and automated email notifications.

The workflow of the system was:

1. Users upload insurance claim documents.
2. OCR extracts text from uploaded files.
3. NLP preprocesses and cleans the extracted data.
4. BioGPT embeddings generate contextual vector representations during preprocessing.
5. SVM and XGBoost models perform fraud prediction.

6. SHAP generates explainability reports.
7. Fraud analysis results are displayed on the admin dashboard.
8. The GPT chatbot provides real-time support and explanations.
9. Automated notifications are sent to users.

To ensure deployment readiness and model reusability, I saved trained machine learning models using **.pkl** format and integrated backend APIs for prediction serving.

I served as the **Team Leader** for this project. My responsibilities included:

- model architecture design,
- OCR and NLP integration,
- backend development,
- AI module integration,
- task allocation,
- daily progress tracking,
- deployment coordination,
- and solving technical blockers.

I coordinated with team members regularly and ensured all milestones were completed successfully within the timeline. This role helped me strengthen my leadership, communication, and problem-solving abilities.

Overall, this project significantly improved my knowledge in:

- Machine Learning,
- Explainable AI,
- NLP,
- OCR,
- XGBoost,
- SVM,
- BioGPT embeddings,
- SHAP,
- Flask development,
- Generative AI integration,

- and full-stack system design.

This project gave me real-world experience in building an end-to-end AI-powered healthcare fraud detection system capable of solving practical industry problems using intelligent automation and explainable AI technologies.

NLP Preprocessing Techniques Used in My Project

In my project, after extracting text from insurance documents using OCR, the data was unstructured and noisy. So, I applied several NLP preprocessing techniques before sending the data to the machine learning models.

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1. Data Cleaning

What is Data Cleaning?

Data cleaning is the process of removing unnecessary, incorrect, or inconsistent data from the extracted text.

Why was it needed in my project?

Insurance documents contained:

- special symbols,
- unwanted spaces,
- spelling inconsistencies,
- duplicate text,
- OCR extraction noise,
- irrelevant characters.

Example OCR Output:

P@tient N4me : R@mesh Kumar ###

Claim Amt : $$$50000

After Cleaning:

Patient Name : Ramesh Kumar

Claim Amount : 50000

How I used it in my project

I cleaned:

- unwanted symbols,
- HTML characters,
- punctuation noise,
- extra spaces,
- duplicate entries.

This improved data quality before feature extraction and model training.

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2. Tokenization

What is Tokenization?

Tokenization is the process of splitting text into smaller meaningful units called tokens.

These tokens can be:

- words,
- sentences,
- or subwords.

Example

Original Sentence:

Patient admitted for heart surgery

After Tokenization:

["Patient", "admitted", "heart", "surgery"]

Why was it needed in my project?

Machine learning models cannot directly understand raw sentences.

So tokenization helped convert medical text into smaller analyzable units for:

- feature extraction,

- fraud pattern analysis,
- medical term processing.

How I used it in my project

I tokenized:

- diagnosis descriptions,
- treatment summaries,
- hospital records,
- claim explanations.

This helped the ML models analyze claim-related keywords effectively.

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3. Stop-Word Removal

What are Stop Words?

Stop words are commonly used words that do not add important meaning to prediction tasks.

Examples:

- the
- is
- and
- of
- was
- are

Example

Before:

The patient was admitted in the hospital

After Stop-Word Removal:

["patient", "admitted", "hospital"]

Why was it needed in my project?

Insurance documents contain many unnecessary common words.

Removing stop words:

- reduced noise,
- reduced feature size,
- improved processing speed,
- improved model efficiency.

How I used it in my project

I removed stop words from:

- medical descriptions,
- claim summaries,
- hospital reports.

This helped focus only on important fraud-related keywords.

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4. Normalization

What is Normalization in NLP?

Normalization converts text into a standard and consistent format.

Example

Before:

PATIENT admitted!!!

patient Admitted

After Normalization:

patient admitted

What normalization included in my project

I performed:

- lowercase conversion,
- removing extra spaces,
- removing special characters,
- standardizing text format.

Why was it needed?

Medical documents came from different hospitals and formats.

Normalization helped:

- maintain consistency,
- reduce duplicate patterns,
- improve model understanding.

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5. Feature Extraction

What is Feature Extraction?

Feature extraction is the process of identifying important information from text and converting it into numerical features that ML models can understand.

Machine learning models work only with numbers, not raw text.

Example Features in My Project

From insurance claims, I extracted:

- diagnosis keywords,
- treatment type,
- claim amount,
- repeated hospitalization patterns,
- suspicious billing patterns,
- patient history,
- hospital information.

How I used Feature Extraction

After preprocessing:

- important medical keywords were identified,
- BioGPT embeddings generated contextual vectors,
- numerical features were created,
- features were passed into SVM and XGBoost models.

Complete NLP Workflow in My Project

Step 1

OCR extracted text from uploaded insurance documents.

↓

Step 2

Data cleaning removed unwanted noise.

↓

Step 3

Tokenization split text into meaningful tokens.

↓

Step 4

Stop-word removal removed unnecessary words.

↓

Step 5

Normalization standardized the text.

↓

Step 6

Feature extraction generated meaningful ML features.

↓

Step 7

BioGPT embeddings created contextual vector representations.

↓

Step 8

SVM and XGBoost models performed fraud prediction.

Why These Steps Were Important

These preprocessing techniques helped:

- improve text quality,
- reduce noise,
- improve model accuracy,
- enhance fraud detection capability,
- reduce false positives,
- and improve contextual understanding of insurance claims.

Without preprocessing, the ML models would not accurately understand medical documents or fraud patterns.

How OCR Text Was Extracted and Processed in My Project

In my project, users uploaded scanned insurance documents such as:

- medical bills,
- prescriptions,
- discharge summaries,
- and claim forms.

These documents were mostly image-based PDFs or scanned images, so machine learning models could not directly understand them.

So first, I used **OCR (Optical Character Recognition)** to extract text from the uploaded documents.

Step 1: Document Upload

The user uploads:

- PDF,
- JPG,
- PNG,
- or scanned medical document

through the Flask web application.

Example:

medical_claim.pdf

Step 2: OCR Processing

I used OCR libraries like:

- Tesseract OCR
- EasyOCR

to scan the uploaded document and extract text.

Example OCR Extracted Output:

P@tient N4me : R@mesh Kumar ###

Claim Amt : $$$50000

This output contained:

- OCR noise,
- special characters,
- spelling distortions,
- extra symbols.

Because scanned documents are sometimes blurry or low quality, OCR may produce incorrect characters.

Step 3: Data Cleaning

After OCR extraction, I applied preprocessing and cleaning techniques.

What I cleaned:

- special characters
- symbols
- unwanted punctuation
- duplicate spaces
- OCR noise

Example Cleaning Process

Before Cleaning:

P@tient N4me : R@mesh Kumar ###

Claim Amt : $$$50000

Cleaning Logic

Remove Special Characters

Removed:

- @
-
- $
- extra punctuation

Normalize Text

Converted inconsistent formatting into standard text.

Example:

N4me → Name

Amt → Amount

After Cleaning:

Patient Name : Ramesh Kumar

Claim Amount : 50000

Step 4: Tokenization

Then I split the cleaned text into tokens.

Example:

["Patient", "Name", "Ramesh", "Kumar", "Claim", "Amount", "50000"]

This helped the ML model process words individually.

Step 5: Stop-Word Removal

Unnecessary words were removed.

Example:

"The patient was admitted in hospital"

↓

["patient", "admitted", "hospital"]

This reduced noise in the dataset.

Step 6: Feature Extraction

Then I extracted important fraud-related features like:

- patient name
- hospital details
- claim amount
- diagnosis
- treatment type
- repeated claim patterns

Step 7: BioGPT Embeddings

After preprocessing, I used **BioGPT embeddings** to convert medical text into contextual vector representations.

This helped the system better understand:

- medical terminology,
- diagnosis relationships,
- healthcare context.

Step 8: ML Prediction

Finally:

- SVM analyzed text-based fraud patterns
- XGBoost analyzed structured numerical features

Then the system predicted whether the claim was:

- fraudulent
- or legitimate.

Why This Process Was Important

Without preprocessing:

- OCR text would contain noise,
- ML models would misinterpret data,
- prediction accuracy would decrease.

These preprocessing steps improved:

- text quality,
- feature extraction,
- fraud detection accuracy,
- and overall system reliability.

How Tesseract OCR and EasyOCR Work

In my project, I used OCR (Optical Character Recognition) to extract text from scanned insurance documents such as:

- medical bills,

- prescriptions,
- claim forms,
- discharge summaries.

OCR converts image-based text into machine-readable text so that NLP and ML models can process it.

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1. Tesseract OCR

What is Tesseract OCR?

Tesseract OCR is an open-source OCR engine developed by Google.

It is used to detect and extract printed text from:

- images,
- scanned PDFs,
- documents.

How Tesseract OCR Works

Step 1: Input Image

The system receives:

- scanned image,
- PDF,
- or document.

Example:

- medical bill image.

Step 2: Image Preprocessing

Before extracting text, Tesseract preprocesses the image.

It performs:

- grayscale conversion,
- noise removal,

- thresholding,
- image sharpening.

This improves text visibility.

Step 3: Text Detection

Tesseract identifies:

- text regions,
- characters,
- words,
- lines.

It separates text from background.

Step 4: Character Recognition

The OCR engine compares detected characters with trained language patterns and predicts characters.

Example:

P@tient N4me

may become:

Patient Name

Step 5: Text Output

Finally, extracted text is returned as machine-readable text.

Example:

Patient Name : Ramesh Kumar

Claim Amount : 50000

Why I Used Tesseract OCR

Advantages:

- open source,
- easy integration with Python,
- supports multiple languages,
- good for structured printed documents.

In my project, I used it for:

- scanned medical forms,
- insurance bills,
- printed healthcare documents.

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2. EasyOCR

What is EasyOCR?

EasyOCR is a deep learning–based OCR library.

Unlike Tesseract, EasyOCR uses neural networks for text detection and recognition.

How EasyOCR Works

Step 1: Image Input

The uploaded document image is passed into EasyOCR.

Step 2: Text Detection

EasyOCR uses deep learning models to detect text regions inside the image.

It can identify:

- curved text,
- rotated text,
- low-quality text,
- handwritten-like text.

Step 3: Character Recognition

EasyOCR uses neural networks to recognize characters and words.

It predicts text contextually using learned patterns.

Step 4: Extracted Text Output

Example:

Claim Amount : 50000

Hospital : Apollo

Why I Used EasyOCR

EasyOCR worked better for:

- low-quality scans,
- complex medical documents,
- noisy images,
- irregular formatting.

It provided better flexibility than traditional OCR in some cases.

Difference Between Tesseract OCR and EasyOCR

Feature	Tesseract OCR	EasyOCR
Type	Traditional OCR	Deep Learning OCR
Speed	Faster	Slightly slower
Accuracy on clean text	Good	Good
Accuracy on noisy images	Moderate	Better
Handwritten/rotated text	Limited	Better
Dependency	Rule-based OCR	Neural network–based

How I Used OCR in My Project Workflow

Step 1

User uploads insurance document

↓

Step 2

OCR extracts text using:

- Tesseract OCR
- or EasyOCR

↓

Step 3

Extracted text sent to NLP preprocessing

↓

Step 4

Data cleaning, tokenization, normalization performed

↓

Step 5

Features extracted and passed to:

- SVM
- XGBoost

↓

Step 6

Fraud prediction generated

Why OCR Was Important in My Project

Without OCR:

- ML models cannot understand scanned documents,
- manual typing would be required,
- automation would not be possible.

OCR helped automate:

- document reading,
- data extraction,

- fraud analysis,
- and claim verification workflow.

Why Did You Use SMOTE?

Interview Answer

In our project, the dataset was highly imbalanced because genuine insurance claims were much higher than fraudulent claims. If we trained the model directly on this data, the model would become biased toward the majority class and might fail to identify fraud cases correctly.

To solve this issue, I used **SMOTE (Synthetic Minority Oversampling Technique)**. SMOTE generates synthetic samples for the minority class instead of simply duplicating existing fraud records.

This helped balance the dataset and improved the model's ability to learn fraud patterns effectively. As a result, metrics like **Recall** and **F1-score** improved significantly, which is very important in fraud detection systems because missing fraudulent claims can lead to major financial losses.

What is SMOTE?

Simple Explanation

SMOTE is a data balancing technique used when:

- one class has very few records,
- and another class has many records.

Example:

- Genuine claims = 90%
- Fraud claims = 10%

This creates **class imbalance**.

SMOTE creates artificial fraud samples to balance the dataset.

How SMOTE Works

Step-by-Step

Before SMOTE

Claim Type	Count
Genuine	90,000
Fraud	10,000

The model may predict most claims as genuine.

After SMOTE

SMOTE creates synthetic fraud records.

Claim Type	Count
Genuine	90,000
Fraud	90,000

Now the model learns fraud patterns properly.

Explain NLP Preprocessing Steps

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1. Tokenization

What is Tokenization?

Tokenization splits text into smaller meaningful units called tokens.

Example

Before:

Patient admitted for surgery

After:

["Patient", "admitted", "surgery"]

How I Used It

In my project, tokenization helped split:

- medical reports,
- diagnosis details,
- claim descriptions

into analyzable words for machine learning models.

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2. Normalization

What is Normalization?

Normalization converts text into a standard format.

Example

Before:

PATIENT admitted!!!

After:

patient admitted

What I Did

I performed:

- lowercase conversion,
- special character removal,
- extra-space removal,
- text standardization.

This improved consistency in the dataset.

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3. Stop-Word Removal

What are Stop Words?

Stop words are common words that do not contribute much meaning.

Examples:

- the

- is
- and
- was

Example

Before:

The patient was admitted in hospital

After:

["patient", "admitted", "hospital"]

Why I Used It

This reduced noise and improved model efficiency by focusing only on important medical and fraud-related words.

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4. Feature Extraction

What is Feature Extraction?

Feature extraction converts processed text into meaningful numerical information that ML models can understand.

Features Used in My Project

I extracted:

- diagnosis keywords,
- treatment details,
- claim amount,
- repeated hospitalization,
- suspicious billing patterns,
- hospital information.

These features were passed into:

- SVM

- XGBoost

for fraud prediction.

Important Points to Explain in Interview

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1. Problem Statement

Insurance fraud causes huge financial losses, and manual claim verification is slow and error-prone. So, the goal of our project was to build an AI-powered automated fraud detection system for health insurance claim verification.

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2. OCR Workflow

Users upload scanned insurance documents. OCR extracts machine-readable text from uploaded PDFs and images. This extracted text is then passed to the NLP preprocessing pipeline.

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3. NLP Preprocessing

After OCR extraction, I applied:

- data cleaning,
- tokenization,
- stop-word removal,
- normalization,
- and feature extraction
  to convert unstructured medical text into meaningful ML-ready data.

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4. SVM + XGBoost

I used a hybrid ML approach using SVM and XGBoost.

SVM worked well for text-based classification, while XGBoost handled structured numerical data efficiently and improved scalability and prediction accuracy.

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5. BioGPT Embeddings

During preprocessing, I used BioGPT embeddings to generate contextual vector representations from healthcare-related text. This improved the system's understanding of medical terminology and claim descriptions.

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6. SHAP Explainability

I implemented SHAP Explainable AI to display which features influenced fraud predictions. This improved transparency and helped admins understand why a claim was flagged as fraudulent.

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7. Flask Integration

I built a Flask-based full-stack web application with:

- secure login/signup,
- claim upload system,
- admin dashboard,
- fraud analysis reports,
- chatbot integration,
- and automated email notifications.

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8. Final Accuracy

The hybrid AI model achieved approximately **98.48% accuracy** with strong Precision, Recall, F1-score, and ROC-AUC performance, making the system reliable for fraud detection and claim verification.

Precision, Recall, and F1-Score — Clear Interview Explanation

In fraud detection projects, Accuracy alone is not enough because the dataset is usually imbalanced.

Example:

- Genuine claims = 95%
- Fraud claims = 5%

Even if the model predicts everything as genuine, it may still show high accuracy, but it completely fails to detect fraud.

So, metrics like:

- Precision
- Recall
- F1-score

become very important.

First Understand These Terms

Term	Meaning
TP (True Positive)	Fraud claim correctly predicted as fraud
TN (True Negative)	Genuine claim correctly predicted as genuine
FP (False Positive)	Genuine claim wrongly predicted as fraud
FN (False Negative)	Fraud claim wrongly predicted as genuine

Example

Suppose:

Actual Claim	Predicted Claim
Fraud	Fraud → TP
Genuine	Fraud → FP
Fraud	Genuine → FN
Genuine	Genuine → TN

---

1. Precision

Formula

Precision = \frac{TP}{TP + FP}

What Precision Means

Precision tells:

Out of all claims predicted as fraud, how many were actually fraudulent?

Simple Real-Time Example

Suppose:

- Model predicted 100 claims as fraud
- Out of those:
    - 90 were actually fraud
    - 10 were genuine claims wrongly flagged

Then:

TP FP

90 10

Precision:

90 / (90 + 10) = 0.90

Precision = 90%

Why Precision Matters in Fraud Detection

High Precision means:

- fewer genuine claims are wrongly blocked,
- fewer unnecessary manual reviews,
- better customer experience.

Interview Answer for Precision

Precision measures how many claims predicted as fraudulent were actually fraudulent. In fraud detection systems, high precision helps reduce false alarms and unnecessary claim investigations.

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2. Recall

Formula

Recall = \frac{TP}{TP + FN}

What Recall Means

Recall tells:

Out of all actual fraud claims, how many were correctly identified by the model?

Example

Suppose:

- Total actual fraud claims = 100
- Model correctly identified 95 fraud claims
- Missed 5 fraud claims

TP FN

95 5

Recall:

95 / (95 + 5) = 0.95

Recall = 95%

Why Recall Is Very Important in Fraud Detection

Low Recall means:

- fraud claims are missed,
- financial loss increases,
- fraudulent users escape detection.

In fraud detection:

Missing fraud is more dangerous than wrongly flagging a genuine claim.

So Recall becomes extremely important.

Interview Answer for Recall

Recall measures how many actual fraudulent claims were correctly identified by the model. In fraud detection systems, high recall is very important because missing fraudulent claims can lead to major financial losses.

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3. F1-Score

Formula

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

What F1-Score Means

F1-score balances:

- Precision
- Recall

It gives a single performance score.

Why F1-score Is Important

Sometimes:

- Precision is high
- Recall is low

or:

- Recall is high
- Precision is low

F1-score helps evaluate the overall balance.

Example

Suppose:

- Precision = 90%
- Recall = 80%

F1-score gives combined performance.

Why F1-score Was Important in My Project

Our fraud dataset was imbalanced.

So relying only on accuracy was not reliable.

F1-score helped evaluate:

- fraud detection quality,
- balance between false positives and false negatives.

Interview Answer for F1-score

F1-score is the harmonic mean of Precision and Recall. It balances both metrics and is very useful in fraud detection problems where the dataset is imbalanced and both false positives and false negatives are important.

Very Important Interview Question

Which is More Important in Fraud Detection: Precision or Recall?

Best Answer

In fraud detection, Recall is usually more important because missing fraudulent claims can cause major financial losses. However, maintaining a good balance between Precision and Recall is also necessary to reduce unnecessary investigations. That's why F1-score becomes an important evaluation metric.

Simple Memory Trick

Metric	Simple Meaning
Precision	How correct are fraud predictions?
Recall	How many frauds did we catch?
F1-score	Balance between Precision and Recall

Your Project Metrics Explanation

From your project:

Metric	Value
Precision	0.98–0.99
Recall	0.97–0.99
F1-score	0.98
Accuracy	98.48%

How to Explain These Results

The model achieved balanced performance across all evaluation metrics. High precision reduced false fraud alerts, high recall ensured most fraudulent claims were detected, and the strong F1-score showed balanced and reliable fraud detection performance.

What is Linearly Separable?

Simple Meaning

Data is called **linearly separable** if we can separate two classes using a straight line.

Example:

- Fraud claims on one side
- Genuine claims on another side

using a single straight boundary.

Example of Linear Separation

Imagine:

Fraud Claims	Genuine Claims
Left side	Right side

A straight line can separate them.

Example:

Fraud | Genuine

Fraud | Genuine

Fraud | Genuine

The vertical line separates both classes clearly.

This is called:

Linearly Separable Data

What is Non-Linear Separation?

In real-world fraud detection:

- fraud patterns are complex,
- overlapping,
- irregular.

A straight line cannot separate them properly.

Example:

Genuine Fraud Genuine

Fraud Genuine Fraud

Genuine Fraud Genuine

Here:

- classes are mixed,
- no straight line can separate them.

This is called:

Non-Linearly Separable Data

Why Fraud Detection is Non-Linear

Insurance fraud depends on many complex relationships:

- billing amount,
- diagnosis,
- repeated claims,
- hospital patterns,
- patient history,

- text descriptions.

Fraud does not follow a simple straight-line pattern.

So we need more advanced separation techniques.

What Does Kernel Do in SVM?

The **kernel trick** helps SVM handle non-linear data.

Instead of using a straight line:

- the kernel transforms data into higher dimensions,
- where separation becomes easier.

Simple Explanation

Kernel helps SVM:

create curved decision boundaries instead of straight lines.

This helps capture:

- complex fraud behavior,
- hidden patterns,
- overlapping classes.

Real Interview Explanation

Insurance fraud patterns are rarely linearly separable because fraud behavior is complex and depends on multiple features. The kernel trick helps SVM transform data into higher-dimensional space and separate complex fraud patterns using non-linear boundaries.

Why RBF Kernel Was Used

RBF = Radial Basis Function

$$K(x,x') = \exp(-\gamma ||x-x'||^2)$$

Why RBF Kernel is Suitable

Reason	Benefit
Handles non-linear data	Captures complex fraud patterns
Works with text features	Good for NLP embeddings
Flexible boundary	Improves prediction accuracy
Handles overlapping classes	Better fraud classification

Simple Real-Time Analogy

Imagine:

- straight road = linear separation
- curved road = non-linear separation

Fraud detection needs curved boundaries because fraud behavior is complex.

RBF kernel helps create those curved boundaries.

Interview Answer (Short Version)

Fraud patterns are complex and cannot be separated using a simple straight line. So, I used the RBF kernel in SVM because it helps capture non-linear relationships by creating flexible decision boundaries in higher-dimensional space.

Interview Answer (Technical Version)

The kernel trick allows SVM to compute similarities between data points in higher-dimensional space without explicitly transforming the data. Since insurance fraud patterns are non-linear and overlapping, the RBF kernel helped create non-linear decision boundaries and improved classification performance.

. **How Did You Choose the Algorithm?**

I chose the algorithm based on the problem requirements, data size, accuracy needs, and performance considerations.

First, I analyzed the type of problem — whether it was classification, regression, clustering, or prediction. Then I compared different algorithms based on factors like accuracy, training time, interpretability, and scalability.

For example, if the dataset was structured and needed high accuracy, I considered algorithms like Random Forest or XGBoost. If interpretability was important, I preferred Logistic Regression or Decision Trees.

I also evaluated the algorithm using metrics such as accuracy, precision, recall, F1-score, or RMSE depending on the project. After testing multiple models, I selected the one that gave the best balance between performance and efficiency.